

Simon Says Memory Game with Arduino

Overview

This project implements a "Simon Says"-style memory game using an Arduino. A sequence of 4 LEDs lights up randomly, and the user must reproduce it using 4 directional buttons. Each successful try increases the difficulty level.

Hardware

- Arduino Uno
- LEDs: Red (9), Green (8), White (7), Blue (10)
- Buttons: Left (3), Right (4), Up (5), Down (6)
- Resistors, breadboard, jumper wires

Functionality

- Generates a 6-step random LED sequence.
- Player reproduces it using buttons.
- Difficulty level adjusts LED display speed.
- 5-second timeout ends current round.

Code Summary

- Uses Serial to set difficulty.
- Stores and verifies input sequences.
- Implements timeout using `millis()`.
- Provides result via Serial monitor.

Simulation

Tinkercad Circuit Online: <https://www.tinkercad.com/things/iUHdDMSBkBK-frantic-jofo-waasa>